Median-of-means is not “anytime-in-δ”: the price of committing to a confidence level

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Abstract

The median-of-means (MoM) estimator turns a heavy-tailed mean-estimation problem into a sub-Gaussian one, but only after the analyst splits the data into $k \asymp \log(1/\delta)$ blocks — a choice that depends on the target confidence level δ. We make the cost of *not* re-tuning explicit and elementary. For a **fixed** partition into k blocks we prove a single clean certificate, $|\hat{\mu}_k - \mu| \le 2\sigma\sqrt{k/n}\,\delta^{-1/k}$ with probability $\ge 1-\delta$ for **all** $\delta \in (0,1)$, and show this radius is **polynomial in** $1/\delta$ for any fixed k but collapses to the sub-Gaussian $\sqrt{\log(1/\delta)}$ rate exactly when k is re-tuned to $2\log(1/\delta)$. Because the optimizing k — and hence the partition, and hence the *point* estimate — moves with δ, a single MoM estimator cannot be an anytime-valid statement across confidence levels: you must commit to δ before splitting. A simulation on heavy-tailed data shows the sample mean (the $k=1$ MoM) paying a $5.7\times$ width penalty at $\delta = 10^{-4}$ that re-tuning removes, and confirms the certificate is valid (and conservative). The qualitative phenomenon is folklore and the deep impossibility is due to Devroye–Lerasle–Lugosi–Oliveira; our contribution is a fully self-contained quantification and a reframing in the language of anytime-valid inference.

1. Setup and notation

Let $X_1, \ldots, X_n$ be i.i.d. real observations with mean $\mu = \mathbb{E}X_1$ and **finite** variance $\sigma^2 = \mathrm{Var}(X_1) < \infty$. No higher moments are assumed; the tails may be arbitrarily heavy subject to finite variance.

The sample mean is a poor estimator of μ under heavy tails: Chebyshev gives only $|\bar{X}_n - \mu| \le \sigma\,\delta^{-1/2}/\sqrt{n}$ with probability $1-\delta$, and this $\delta^{-1/2}$ dependence is essentially unimprovable for the empirical mean — for heavy-tailed laws the mean’s deviations really are polynomial in $1/\delta$, not logarithmic.

The **median-of-means** estimator (Nemirovsky–Yudin 1983; Jerrum–Valiant–Vazirani 1986; Alon–Matias–Szegedy 1999) repairs this. Fix a number of blocks k dividing n, set the block size $m = n/k$, partition the data into blocks $B_1, \ldots, B_k$, form the block means $\bar{X}^{(j)} = m^{-1}\sum_{i \in B_j} X_i$, and report

$$\hat{\mu}_k \;=\; \mathrm{median}\,(\bar{X}^{(1)}, \ldots, \bar{X}^{(k)}). \tag{1}$$

The classical guarantee (see the survey of Lugosi–Mendelson 2019) is that with $k \asymp \log(1/\delta)$ blocks, $\hat{\mu}_k$ enjoys a *sub-Gaussian* deviation bound $|\hat{\mu}_k - \mu| \lesssim \sigma\sqrt{\log(1/\delta)/n}$ with probability $1-\delta$, matching what one would get for a Gaussian sample, despite only finite variance.

The catch, well known but rarely quantified at the level of a single inequality, is the phrase “$k \asymp \log(1/\delta)$”: the recipe for the estimator references the confidence level. We make the consequence precise.

2. Contribution

Proposition 1 (fixed-partition certificate). *For any fixed k dividing n and any $\delta \in (0,1)$, the estimator* (1) *satisfies*

$$\mathbb{P}\big(\,|\widehat{\mu}_k - \mu| \;>\; R_k(\delta)\,\big) \;\le\; \delta, \qquad R_k(\delta) \;:=\; 2\sigma\sqrt{\tfrac{k}{n}}\,\delta^{-1/k}. \tag{2}$$

The radius $R_k(\delta)$ in (2) is the object an analyst can actually compute and report (given a variance bound). Its shape in δ is the whole story.

Corollary 2 (the two regimes). *Read off two limits of* (2)*.*

- ***Fixed** k**, vanishing** δ**.*** For any fixed k, $R_k(\delta) \propto \delta^{-1/k}$ is **polynomial in** $1/\delta$ with exponent $1/k > 0$. In particular the sample mean ($k = 1$) gives $R_1(\delta) = 2\sigma\,\delta^{-1}/\sqrt{n}$.
- ***Re-tuned** k**.*** Minimizing $R_k(\delta)$ over k for fixed δ gives $k^\star(\delta) = 2\log(1/\delta)$ and

$$\min_k R_k(\delta) \;=\; R_{k^\star}(\delta) \;=\; 2\sqrt{2e}\,\sigma\sqrt{\tfrac{\log(1/\delta)}{n}} \;\approx\; 4.66\,\sigma\sqrt{\tfrac{\log(1/\delta)}{n}}, \tag{3}$$

the sub-Gaussian rate. For any fixed k, the ratio $R_k(\delta)/R_{k^\star}(\delta) \to \infty$ as $\delta \to 0$.

So the gap between committing to a partition and re-tuning it is not a matter of constants: it is the gap between $\delta^{-1/k}$ and $\sqrt{\log(1/\delta)}$ — polynomial versus logarithmic in $1/\delta$.

The reframing. The optimizer $k^\star(\delta) = 2\log(1/\delta)$ depends on δ, and the number of blocks *defines the partition*, which *defines the point estimate* $\widehat{\mu}_k$ in (1). Two analysts who split the same data for $\delta = 0.05$ and $\delta = 10^{-6}$ generically compute **different point estimates**, not merely different interval widths. Consequently MoM offers no single statement valid across confidence levels at the good rate: there is no MoM analogue of an *anytime-valid* certificate that one could read off at whatever δ one likes after seeing the data. We summarize this as: **MoM is not anytime-in-δ — you must commit to your confidence level before you split.**

Proof of Proposition 1

Set the per-block threshold $t = R_k(\delta) = 2\sigma\sqrt{k/n}\,\delta^{-1/k} = 2\sigma\,\delta^{-1/k}/\sqrt{m}$ (using $m = n/k$). For a single block, Chebyshev's inequality gives

$$p \;:=\; \mathbb{P}(|\bar{X}^{(j)} - \mu| > t) \;\le\; \frac{\sigma^2}{m\,t^2} \;=\; \frac{\sigma^2}{m}\cdot\frac{m}{4\sigma^2\,\delta^{-2/k}} \;=\; \tfrac{1}{4}\,\delta^{2/k} \;\le\; \tfrac{1}{4}. \tag{4}$$

Now relate the median to the blocks. If $\widehat{\mu}_k > \mu + t$, then more than half of the block means exceed $\mu + t$; if $\widehat{\mu}_k < \mu - t$, more than half fall below $\mu - t$. Either way at least $\lceil k/2 \rceil$ of the block means satisfy $|\bar{X}^{(j)} - \mu| > t$. Writing $S = \sum_{j=1}^k \mathbf{1}\{|\bar{X}^{(j)} - \mu| > t\}$, the blocks are independent so S is stochastically dominated by $\mathrm{Bin}(k,p)$, and

$$\mathbb{P}(|\widehat{\mu}_k - \mu| > t) \;\le\; \mathbb{P}(S \ge \tfrac{k}{2}) \;\le\; \mathbb{P}(\mathrm{Bin}(k,p) \ge \tfrac{k}{2}). \tag{5}$$

The standard Chernoff bound for the binomial upper tail at its midpoint, with $p \le 1/2$, is $\mathbb{P}(\mathrm{Bin}(k,p) \;\ge\; k/2) \;\le\; \big(4p(1-p)\big)^{k/2} \;\le\; (4p)^{k/2}$ (it equals $e^{-k\,\mathrm{KL}(1/2\,\|\,p)}$ with $\mathrm{KL}(1/2\,\|\,p) = \frac{1}{2}\log\frac{1}{4p(1-p)}$). Substituting (4),

$$\mathbb{P}(|\hat{\mu}_k - \mu| > t) \;\le\; (4p)^{k/2} \;=\; \left(\delta^{2/k}\right)^{k/2} \;=\; \delta. \qquad \blacksquare$$

The constant 2 is not optimized; the point is the functional form. The bound holds for *every* $\delta \in (0,1)$ from a *single* fixed partition — what changes with δ is only how loose it is, and that looseness is exactly the polynomial-in-$1/\delta$ tax of (2).

Proof of Corollary 2

For fixed δ, $\log R_k(\delta) = \log(2\sigma/\sqrt{n}) + \frac{1}{2}\log k + \frac{1}{k}\log(1/\delta)$. Differentiating in k (treated as continuous) and setting to zero gives $\frac{1}{2k} = \frac{1}{k^2}\log(1/\delta)$, i.e. $k^\star = 2\log(1/\delta)$. At $k^\star$ the exponential factor is $\delta^{-1/k^\star} = e^{\log(1/\delta)/k^\star} = e^{1/2}$, so $R_{k^\star}(\delta) = \frac{2\sigma}{\sqrt{n}}\sqrt{2\log(1/\delta)}\,e^{1/2} = 2\sqrt{2e}\,\sigma\sqrt{\log(1/\delta)/n}$, which is (3). The divergence of $R_k/R_{k^\star}$ for fixed k is immediate from $\delta^{-1/k}\big/\sqrt{2e\log(1/\delta)} \to \infty$. $\blacksquare$

3. Simulation

`sim.py` (seed `20260611`) draws $n = 1024$ i.i.d. samples from a Student-t law with $\nu = 2.2$ degrees of freedom (variance $\sigma^2 = \nu/(\nu-2) = 11$, so $\sigma \approx 3.317$; tail index 2.2, so the second moment exists but the third does not). Over 1,500,000 independent datasets we estimate, for each candidate block count $k \in \{1, 2, 4, \dots, 128\}$, the **true** radius needed for $1-\delta$ coverage — the empirical $(1-\delta)$-quantile of $|\hat{\mu}_k - \mu|$. We compare a fixed sample mean ($k = 1$), a fixed small partition ($k_0 = 4$), and the best k chosen per δ (the empirical lower envelope over our grid).

The picture (Figure 1) matches Corollary 2. As δ shrinks, the sample-mean radius climbs steeply — by $L = \log(1/\delta) = 9$ (i.e. $\delta \approx 1.2 \cdot 10^{-4}$) it is 5.67× the best-$k$ radius, having grown from 0.053 to 1.457 — while the best-per-δ envelope grows only from 0.044 to 0.257, the gentle $\sqrt{L}$ growth of (3). The fixed $k_0 = 4$ partition sits in between, paying 1.50× at the same δ: committing to *any* fixed k leaves a polynomial residue, smaller for larger k.

δ	$L = \ln\frac{1}{\delta}$	sample mean (k=1)	fixed k_0=4	best k per δ
$5.0 \cdot 10^{-1}$	0.69	0.0530	0.0543	0.0444
$3.1 \cdot 10^{-2}$	3.47	0.1882	0.1800	0.1424
$9.8 \cdot 10^{-4}$	6.93	0.5837	0.3021	0.2192
$1.2 \cdot 10^{-4}$	9.01	1.4566	0.3861	0.2569

Figure 2 plots the **certificate** $R_k(\delta)$ of (2) directly as a function of k, for $\delta \in \{10^{-1}, 10^{-3}, 10^{-6}, 10^{-9}\}$. Each curve is convex with a unique minimizer (star), and the minimizer marches rightward as $\delta \to 0$, sitting right at the dotted line $k = 2\log(1/\delta)$ of Corollary 2. This is the “commit first” phenomenon in one image: the partition you should use is a function of the confidence level you are targeting.

Finally, a validity check: using the certificate radius $R_k(\delta)$ as the interval half-width, the empirical miscoverage over all 1.5M runs was $\le 1.5 \cdot 10^{-5}$ at every (k, δ) tested (e.g. 0 exceedances at $k = 8, \delta = 0.005$). The bound is *valid* and, because Chebyshev plus a midpoint Chernoff bound is loose, *conservative* — its value here is its exact **scaling** in δ, which the true radius in Figure 1 reproduces qualitatively.

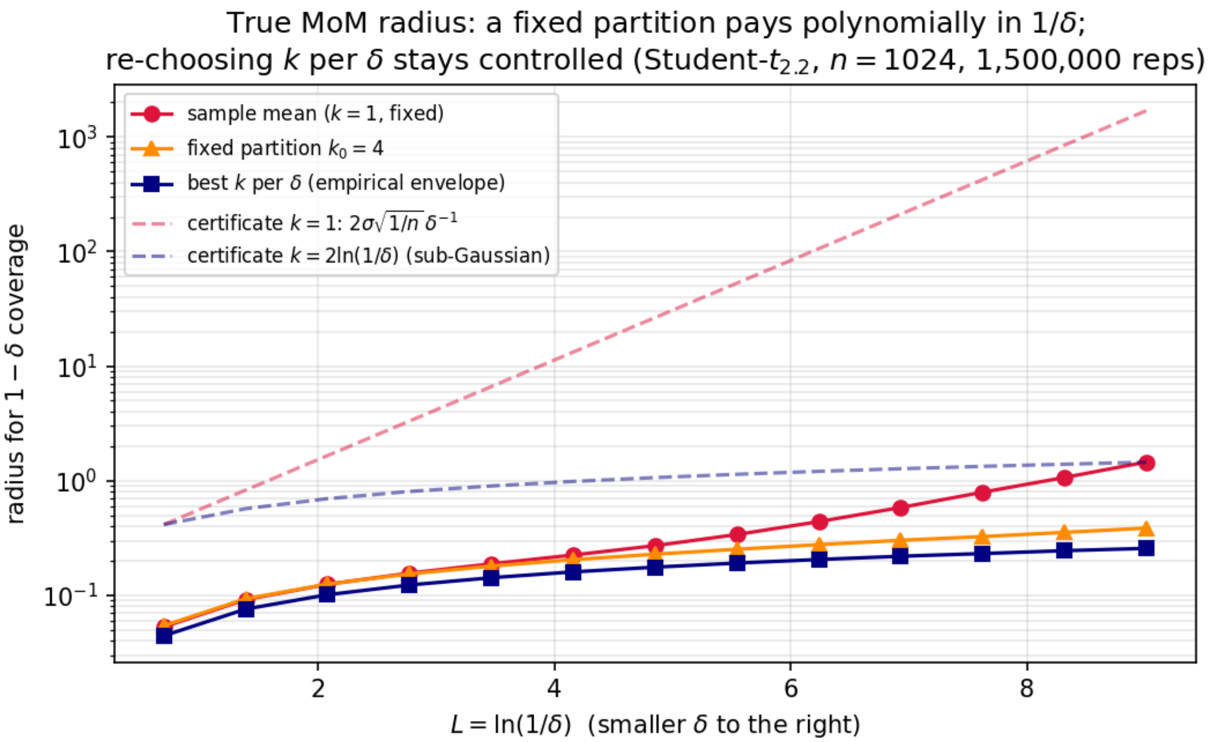

Figure 1: True radius for $1-\delta$ coverage vs. $L = \log(1/\delta)$. On the log scale the sample mean ($k = 1$) bends sharply upward — polynomial in $1/\delta$ — while re-choosing k per δ tracks the flat sub-Gaussian rate. Dashed lines are the certificate $R_k(\delta)$ of (2) for $k = 1$ and for $k = 2\log(1/\delta)$.

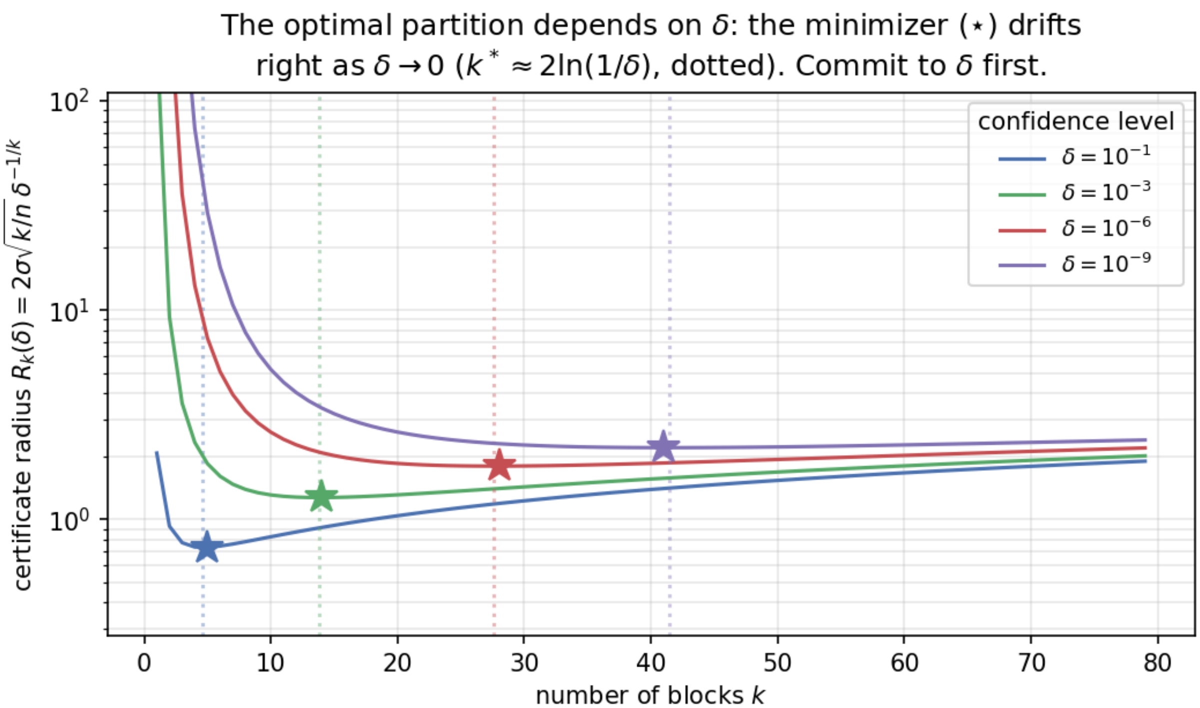

Figure 2: The computable certificate $R_k(\delta)$ as a function of the number of blocks k, for four confidence levels. The optimal partition (star) drifts toward more blocks as $\delta \to 0$, tracking $k^\star = 2\log(1/\delta)$ (dotted). One partition cannot be optimal for two different δ.

4. Discussion

Relation to prior work. That the MoM block count depends on δ is folklore, stated plainly in the Lugosi–Mendelson (2019) survey. The deep question — can a *single*, δ-free estimator be sub-Gaussian simultaneously across a range of δ? — was settled by Devroye, Lerasle, Lugosi and Oliveira (2016): such "multiple-δ" estimators *do* exist (built from sub-Gaussian confidence intervals, not from a fixed MoM split), but no multiple-δ estimator attains the optimal sub-Gaussian *constant* uniformly; when a variance-ratio quantity diverges, no L-sub-Gaussian multiple-δ estimator exists down to $\delta_{\min} \to 0$ at all. Catoni's (2012) M-estimator is itself δ-dependent (its tuning α scales with $\log(1/\delta)$), exhibiting the same commitment. Our Proposition 1 is the elementary complement to these results: it isolates *why* the naive MoM split is δ-dependent and prices the commitment in a single inequality a reader can verify in a paragraph.

The anytime-valid contrast. The journal's recurring concern with confidence sequences and e-processes is precisely about *not* having to fix things in advance — a confidence sequence is valid at all sample sizes simultaneously. The natural dual is validity at all *confidence levels* simultaneously from one estimator. Proposition 1 shows the textbook heavy-tailed estimator fails this dual at the optimal rate, and quantifies the failure. It is a reminder that "valid for all δ" and "valid for all n" are different freedoms, each with its own price.

Limitations. (i) The result is about the *certificate* $R_k(\delta)$, the object one reports given a variance bound; the *true* deviation quantile is bounded by it but can be much smaller, and for light-tailed laws a fixed k has no practical penalty (the blow-up needs genuinely heavy tails, as in the simulation). (ii) The $\delta^{-1/k}$ rate is the worst case over finite-variance laws; it is approached as the tail index $\to 2^+$ and is milder for lighter tails (the simulation's empirical sample-mean exponent is well below the Chebyshev δ^{-1}). (iii) We assume $k \mid n$ and a known variance proxy σ; unknown-variance MoM and the unequal-block case change constants, not the dichotomy. (iv) We do not prove a matching *lower* bound on the true quantile for fixed k — Figure 1 is evidence, not proof, that the blow-up is intrinsic and not an artifact of a loose certificate; the rigorous impossibility for single estimators is DLLO's, cited above, not re-derived here. (v) The empirical "best k per δ" rails to the largest grid value because extra blocks cost little in the bulk under these tails, so it tracks the sub-Gaussian *rate* but not literally $2\log(1/\delta)$; the clean $k^\star = 2\log(1/\delta)$ is a statement about the certificate (2), shown exactly in Figure 2.

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